

DATASCI 507

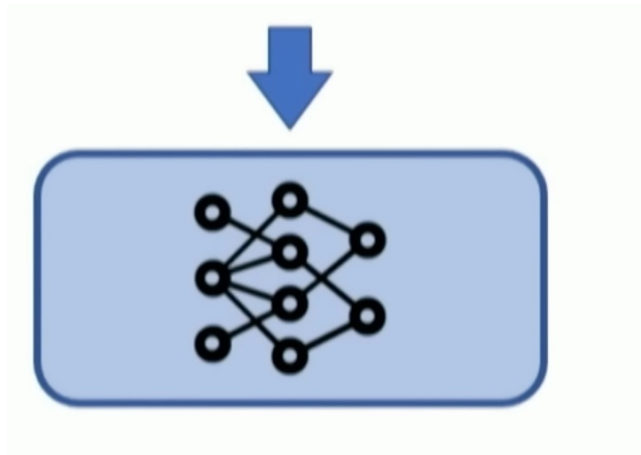
Data Analysis in Python

Week1-1: Syllabus, Installing Python/Jupyter, Intro to Git

Dr. Xian Zhang

Generating Images from Natural Language

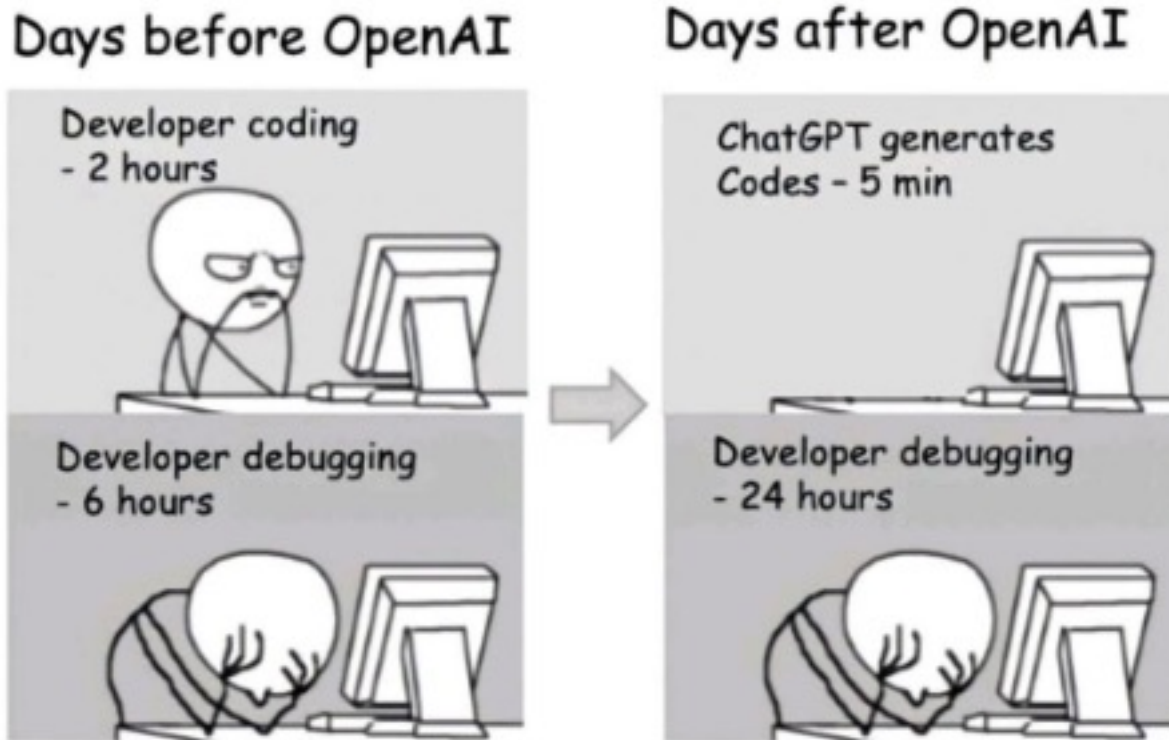
Prompt: “A cat walking a dog at University of Michigan central campus”



Generated by [DALL·E 3](#): able to generate **new data**

Introduction

Big data/ deep learning is revolutionizing so many fields...



A great tool (used properly) for software engineer/data scientist/education

(The ethics guideline: <https://genai.umich.edu/guidance/students>)



[Isaac Gym](#) from Nvidia Isaac Sim

Robotics

Nvidia 2025 CES: <https://www.nvidia.com/en-us/events/ces/>

Introduction

Interacting with large language models



2022

chatGPT3 released



2023

Massive leap, “wow” moment

Introduction

Data-driven physical intelligence



<https://waymo.com/blog/2026/02/ro-on-6th-gen-waymo-driver/>



<https://aibusiness.com/verticals/humanoid-robot-startup-figure-ai-unveils-next-generation-robot>

1. Syllabus

2. Intro to Python and Jupyter

3. Intro to Version Control Systems (git)

Course Goals

1. Establish a broad background in **Python** programming
2. Prepare you for the inevitable coding interview
3. Survey popular **tools** in academia/industry for data analysis
4. Learn how to read documentation and quickly get familiar with new tools.

Tool/programming languages might be obsolete one day...

...But not your ability to learn new frameworks and solve problems!

Scope of STATS 507

Part 1: Introduction to Python

Data types, functions, classes, objects, algorithm thinking.

Part 2: Numerical Computing and Data Visualization

numpy, scipy, scikit-learn, matplotlib, Seaborn

Part 3: Dealing with Structured Data

pandas, SQL, real datasets

Part4: Intro to Deep Learning

PyTorch, Perceptron, Multi-layer perceptron, SGD, CNN, **Agent Coding (new!)**

Prerequisites

Some background on **programming** and **statistics**

Come speak to me if:

- This is your first programming course
- You have never taken a probability or statistics course

This course is probably not for you if:

- You have no programming background

MORE ON THIS LATER

Instructors

Xian Zhang, xianz@umich.edu

Ph.D., Senior Autonomy Engineer II

<https://sites.google.com/view/xian-zhang/home>

My office hours:

Fridays 3:00 PM – 4:30 PM

Zoom: <https://umich.zoom.us/j/8786375189>

Graduate Student Instructors

An Pho, anpho@umich.edu

Statistics PhD student

Siyu Xie, siyuxie@umich.edu

Statistics PhD student

GSI Office hours:

Monday & Wednesdays 10:30 AM – 12:00 AM
Angell Hall 219

Tuesdays & Thursday 11:30 PM – 1:00 PM
Angell Hall 219



Hi, I'm Siyu 🙋

About me

- 2nd year PhD student
- Originally from China
- Research interests: statistical learning, empirical processes

How I can help

- Python concepts & debugging
- Assignments and class materials
- Statistics questions behind the code
- General questions about learning Python / data science

A little outside class

- 🌍 Traveling and photography
- 🎵 Music
- 🗣️ Learning languages

Contact

- ✉️ Email: siyuxie@umich.edu
- 🕒 Office hours:
Tuesday, Thursday 1130-1300

An Pho

Course Website

Canvas

- Read the **syllabus**
- Look at lecture slides and in-class practice
- Download the homework **assignments**
- **Ask questions** about homework assignments
- Find **office hours** times and **zoom** links
- **Submit** homework solutions
- Check your **grades**
- and more!

Getting help

	Canvas discussion board	GSI office hours	my office hours	email GSIs	email me
questions about homework	 *				
questions about lecture / slides / course topics					
questions / concerns about grading			 **		
personal matters; concerns about course; extended illness					

* For questions about homework that cannot be asked without revealing a solution, please ask during GSI office hours rather than through Canvas.

** Please ask the GSIs first about homework grading; come to me if your concern is not resolved.

Course Format

This is a **lecture** format class and **not recorded**. Slides will be uploaded the night before class.

We meet Mondays & Wednesdays, from 8:30 - 10:00 AM in 1200 CHEM. We will end by 9:40 AM for questions and discussions.

I **do not** take attendance.

Textbooks

We recommend the following books, although we will not follow them closely:

The first part of the course is based on *Python for Everybody* by Charles Severance:

<https://www.py4e.com/book.php>

The second part of the course is based on *Understanding Deep Learning* by Simon J.D. Prince:

<https://udlbook.github.io/udlbook/>

Online resources

Another goal of this course is to **get you comfortable reading docs!**

Read and understand what you can, google (chatgpt) terms you don't understand.

- *NumPy quickstart*, <https://numpy.org/devdocs/user/quickstart.html>
- *SciPy tutorial*, <https://docs.scipy.org/doc/scipy/tutorial/index.html>
- *Pyplot tutorial*, <https://matplotlib.org/stable/tutorials/pyplot.html>
- *Pandas tutorial*, https://pandas.pydata.org/pandas-docs/stable/user_guide/
- *Seaborn tutorial*, <https://seaborn.pydata.org/tutorial.html>
- *Scikit-learn tutorial*, <https://scikit-learn.org/stable/tutorial/>
- *PyTorch tutorial*, <https://pytorch.org/tutorials/>

Grading

Final grades will be based on

Assessment	Percentage
Homework (8)	40%
Midterm	30%
Final Project	30%

The distribution of final grades is expected to be similar to what it has been for previous offering of STATS 507:

45% A/A+, 70% A-/A/A+, 98% B- or above.

Homework (40%)

Students enter 507 with a wide variety of programming backgrounds.

The course workload is heavy for some students and light for others.

1. Students with the strongest programming backgrounds may complete each weekly homework assignment in as little as **5 hours**.
2. Many students complete the homework in around **10 hours**.
3. Students with little prior programming experience will likely learn the most; however, they may need to devote as much as **25 hours weekly** to completing the homework assignments!

Homework & late days (40%)

Homework due dates are strict, However,

- You have **seven** “late days” to use over the course of the semester.
- For each late day you spend, you extend the deadline of a homework by 24 hours.
- You may spend multiple late days per homework.
- Once you have turned in your homework you may not spend more late days to turn in your homework again.

The purpose of this late day policy is to enable you to deal with unexpected circumstances (e.g., illness, family emergencies, job interviews) without having to come to me. Of course if dire circumstances arise (e.g., long-term illness that causes you to miss multiple weeks of lecture), please speak with me as promptly as possible. Due to the university grading schedule, you may not use late days to extend the deadline of the last homework assignment.

Don't plagiarize!

- You may discuss homeworks with your fellow students, but the work you submit must be your own.
- You may not share your homework solutions notebook with classmates, and you may not access other students' solutions.
- You must disclose in your homework whom (if anyone) you discussed the homework with.
- Submissions will be checked with the [MOSS Plagiarism Detector](#).
- A zero grade will be given for submissions that contain any plagiarized code.
- All incidents of academic dishonesty will be reported to Rackham, which
- typically administers additional penalties upon conviction.

Midterm (30%)

An **in-person** midterm (closed-book, no cheat-sheet) is scheduled on:

Thursday, 10/15/26 in 1200 CHEM

Evaluation of internalization of core ideas covered in lecture.

Final Projects (30%)

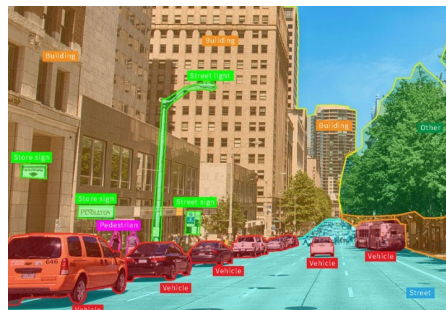
Assuming knowing Python and related tools, the final project for this course is to build ANY end-to-end project with [hugging face](#) where you can use any open-sourced dataset and model (pretrained or fine-tuned).

More details on submission guidelines and evaluation criteria will be provided with the project announcement.

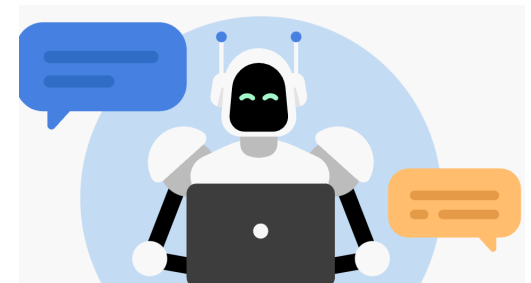
Music generation



Computer Vision



Large Language Models



Questions?

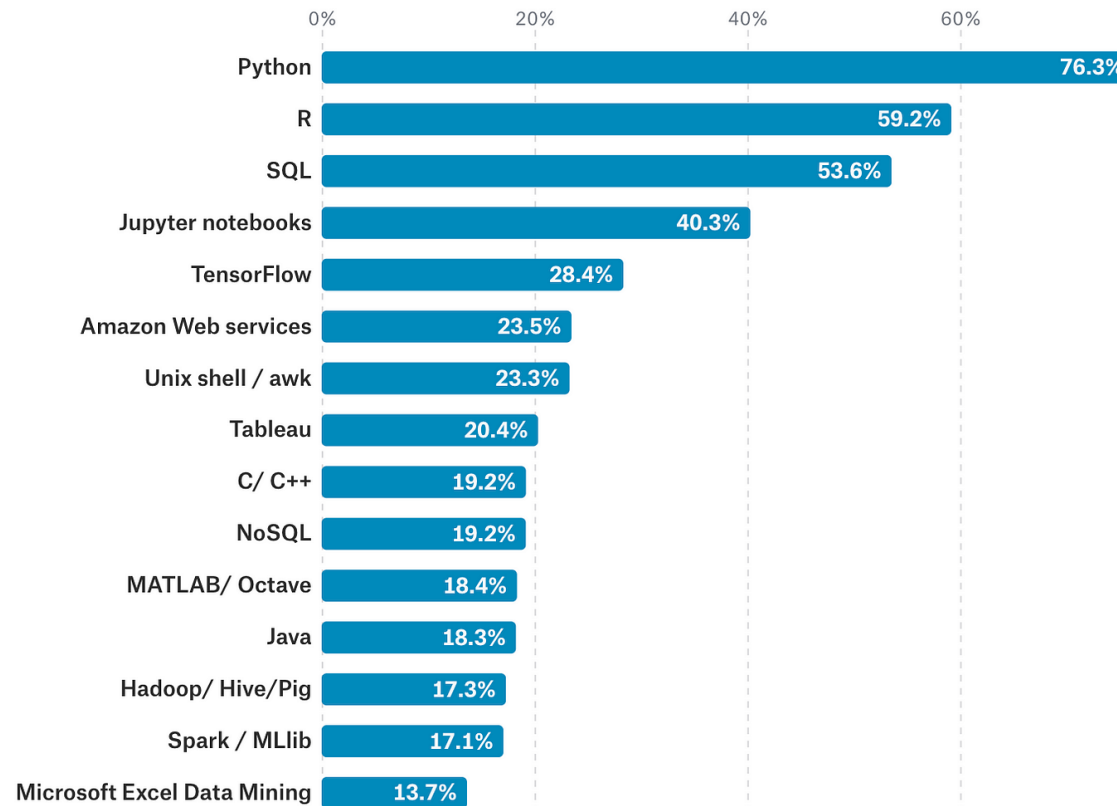
1. Syllabus

2. Intro to Python and Jupyter

3. Intro to version Control

Why Python?

Increasing, Python is the language of data science and general programming.



- Popular
- Easy to learn
- Open-sourced (no license needed)

Data science community Kaggle's annual survey "The State of Data Science& Machine Learning" asks the question "What tools are used at work?" ([reference](#))

What is Python?

Python is a **dynamically typed, interpreted** programming language

- Created by Guido van Rossum and first released in 1991.

Design philosophy: simple, readable

Python is dynamically typed.

Design philosophy: simple, readable

Dynamically typed

In many languages, when you declare a variable, you must specify the variable's **type** (e.g., int, double, Boolean, string). Python does not require this, the type of a variable is defined at **runtime**.

v.s. statically typed, flexible yet more error-prone

Interpreted

Some languages (e.g. C/C++ and Java) are compiled: we write code, from which we get a runnable program via **compilation**. In contrast, Python is **interpreted**: a program, called the **interpreter**, runs our code directly, line by line.

v.s compiled: simple yet slower

An **interpreted** programming language

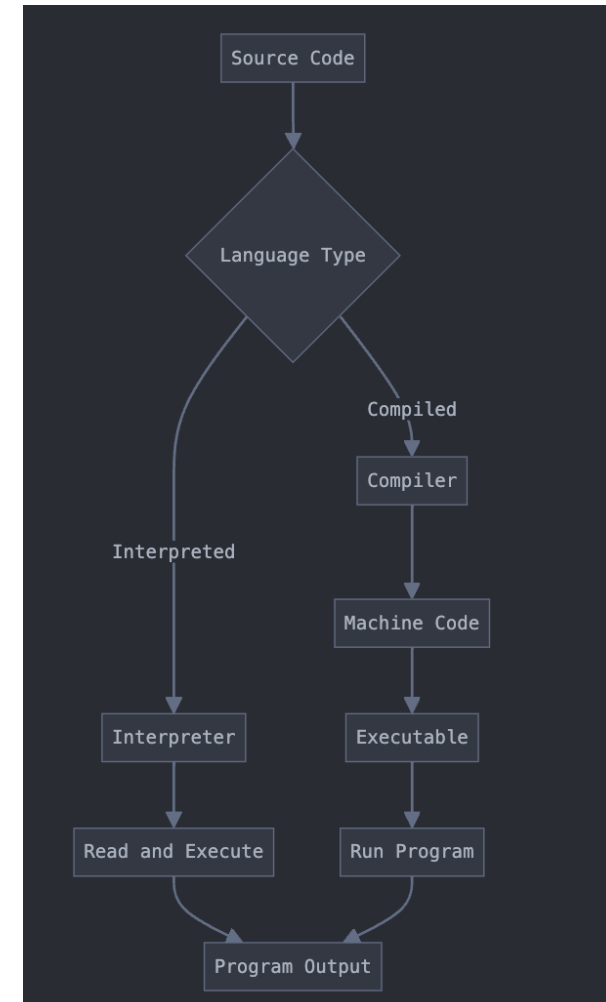
Interpreted

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v.s compiled: simple yet slower

Compiled: Write code -> Compile -> Run executable

Interpreted: Write code -> Run directly with interpreter



Writing and Running Python

Python syntax differs from R, Java, C/C++, MATLAB

- Whitespace delimited
- Limited use of brackets, semicolons, etc.

More on this later...

Running Python



Your HWs and projects must be handed in as **Jupyter** notebooks.

We'll be using Jupyter notebook through Anaconda.

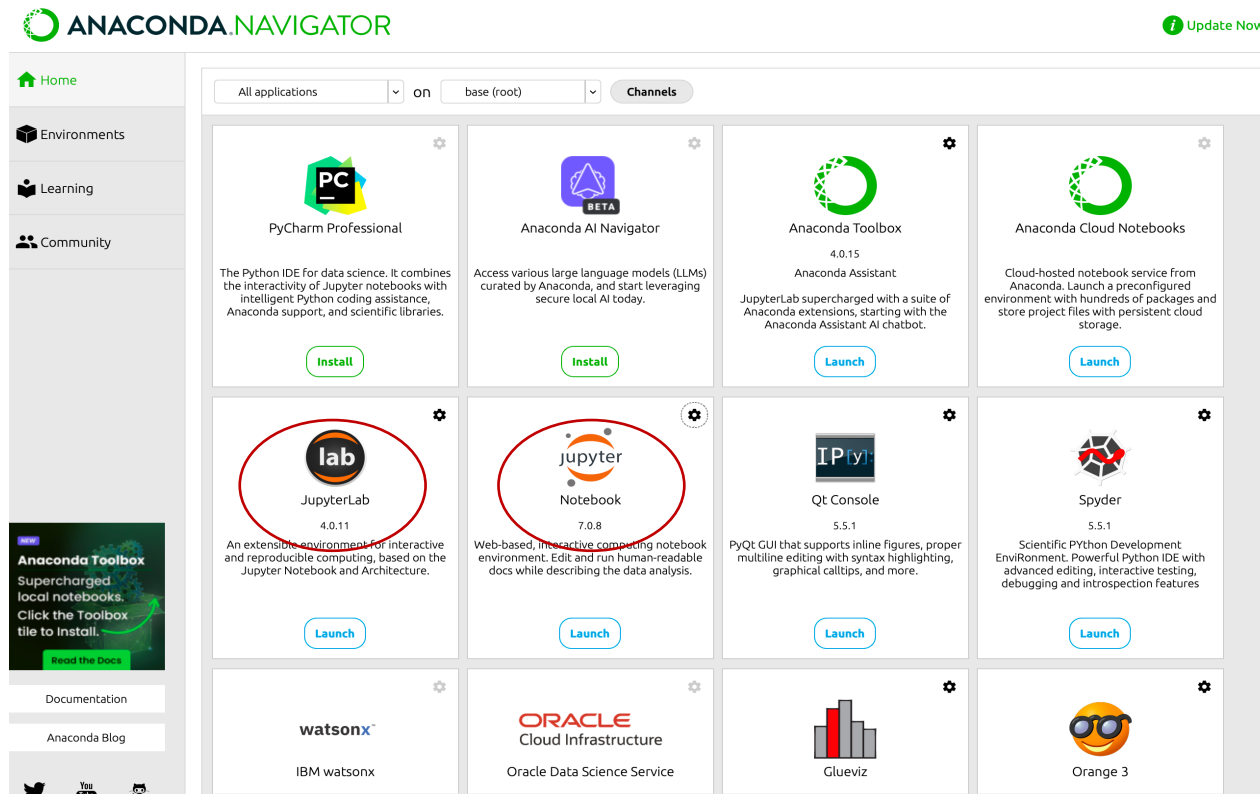
Install Anaconda: <https://docs.anaconda.com/>

Anaconda: A distribution of Python that includes many useful libraries for data science and scientific computing.

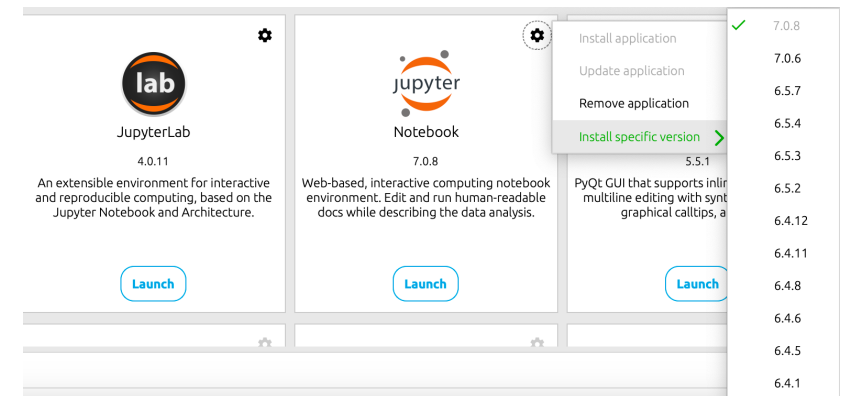
Jupyter Notebook: An interactive web-based environment for creating and sharing documents containing live code, equations, visualizations, and text...

Install Anaconda and launch Jupyter

<https://docs.anaconda.com/anaconda/install/>



Make sure to install the most recent version:



```
import sys
print(f"Python version: {sys.version}")
```

Python version: 3.12.4 | packaged by Anaconda, Inc.

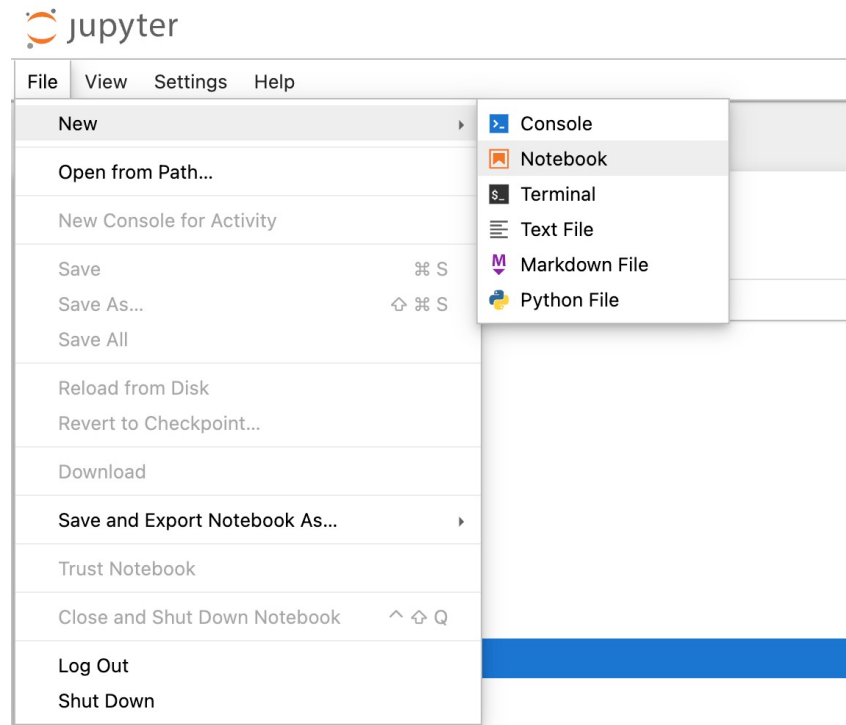
Python in Jupyter

```
xianzhang — jupyter_mac.command — python ◀ jupyter_mac.command — 98x28

File "/opt/anaconda3/lib/python3.12/site-packages/aext_assistant_server/handlers.py", line 1
17, in get
    raise HTTPError(403, reason="missing nucleus_token")
tornado.web.HTTPError: HTTP 403: missing nucleus_token
[W 2024-08-25 15:10:11.970 ServerApp] 403 GET /aext_assistant_server/nucleus_token?1724613011857 (
1f0b9a4c64ce4e1ba1bab40639478aff@::1) 27.13ms referer=http://localhost:8926/tree
[W 2024-08-25 15:10:33.357 ServerApp] Notebook Downloads/week1practice_xian.ipynb is not trusted
[W 2024-08-25 15:10:34.714 ServerApp] wrote error: 'Forbidden'
Traceback (most recent call last):
  File "/opt/anaconda3/lib/python3.12/site-packages/tornado/web.py", line 1790, in _execute
    result = await result
    ^^^^^^^^^^^^^^^^^
File "/opt/anaconda3/lib/python3.12/site-packages/aext_assistant_server/handlers.py", line 1
17, in get
    raise HTTPError(403, reason="missing nucleus_token")
tornado.web.HTTPError: HTTP 403: missing nucleus_token
[W 2024-08-25 15:10:34.714 ServerApp] 403 GET /aext_assistant_server/nucleus_token?1724613034706 (
1f0b9a4c64ce4e1ba1bab40639478aff@::1) 6.94ms referer=http://localhost:8926/notebooks/Downloa/wee
k1practice_xian.ipynb
[W 2024-08-25 15:10:34.767 ServerApp] Notebook Downloads/week1practice_xian.ipynb is not trusted
[I 2024-08-25 15:10:34.923 ServerApp] Kernel started: d9e3f23c-3058-4ae8-9a92-1defbecc86db
0.00s - Debugger warning: It seems that frozen modules are being used, which may
0.00s - make the debugger miss breakpoints. Please pass -Xfrozen_modules=off
0.00s - to python to disable frozen modules.
0.00s - Note: Debugging will proceed. Set PYDEVD_DISABLE_FILE_VALIDATION=1 to disable this validat
ion.
[I 2024-08-25 15:10:35.401 ServerApp] Connecting to kernel d9e3f23c-3058-4ae8-9a92-1defbecc86db.
```

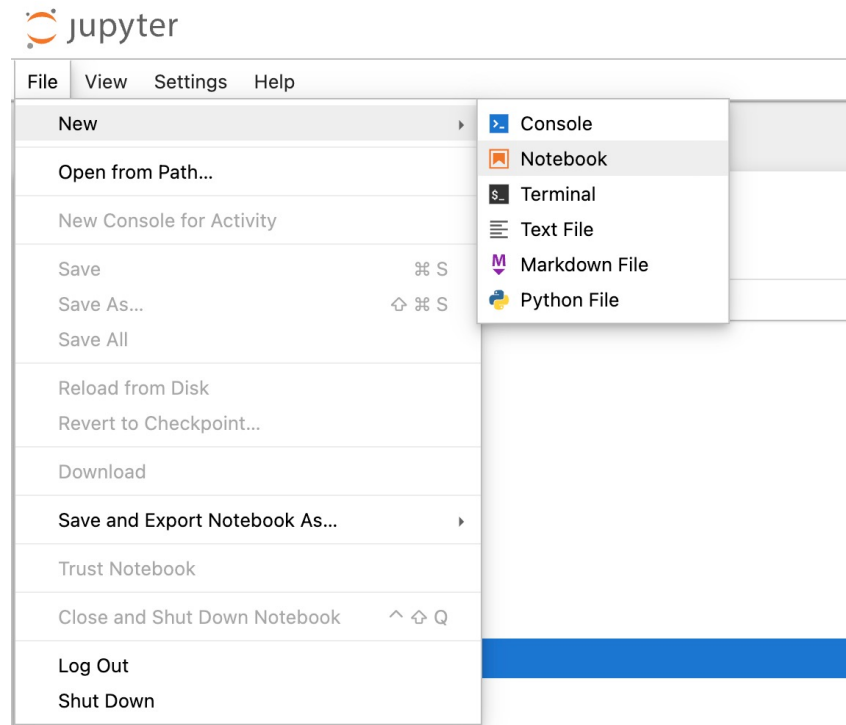
Jupyter provides some information about its startup process, and then...

Python in Jupyter



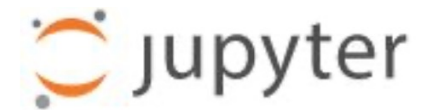
...Jupyter opens a browser window in which you can launch a new notebook or open an existing one...

Python in Jupyter



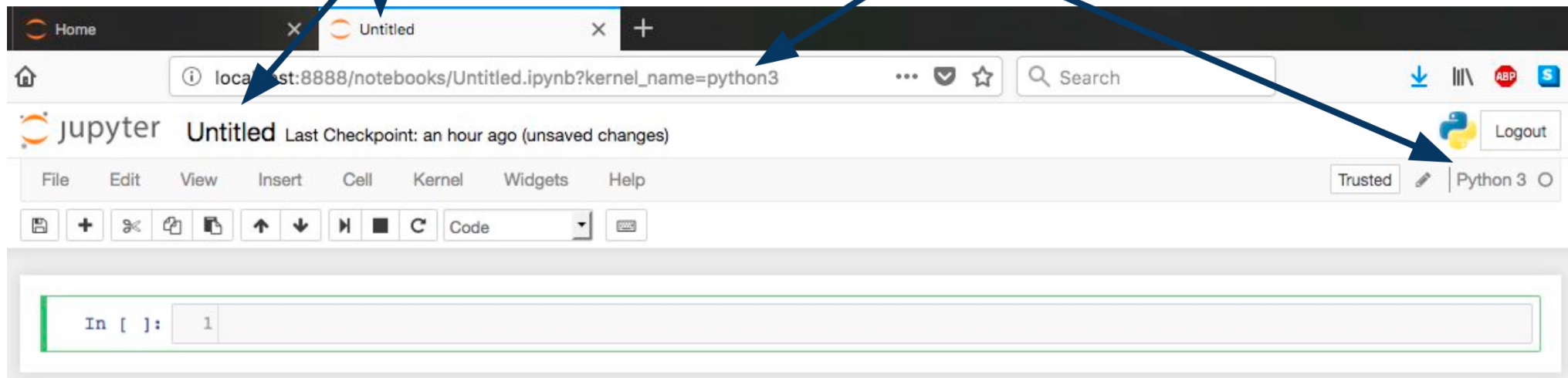
...Jupyter opens a browser window in which you can launch a new notebook or open an existing one...

Python in Jupyter



Notebook doesn't have a title, yet.

Running Python 3



I'll leave it to you to learn about the other features by reading the documentation. For now, the green-highlighted box is most important. That's where we write Python code.

Write code in the highlighted box, then press shift+enter to run the code in that box...

Demo with in-class practice

Resources and support

Example notebook:

<https://nbviewer.jupyter.org/github/jrjohansson/scientific-python-lectures/blob/master/Lecture-4-Matplotlib.ipynb>

Clean, well-organized presentation of code, text and images, in one document

Good tutorials:

<https://www.datacamp.com/tutorial/tutorial-jupyter-notebook>

<https://jupyter-notebook-beginner-guide.readthedocs.io/en/latest/execute.html>

1. Syllabus

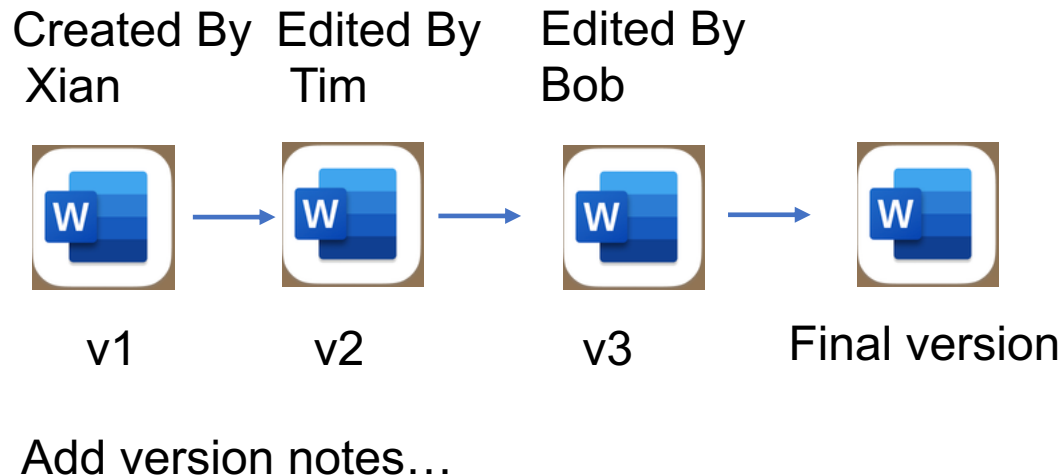
2. Intro to Python and Jupyter

3. Intro to Version Control Systems (git)

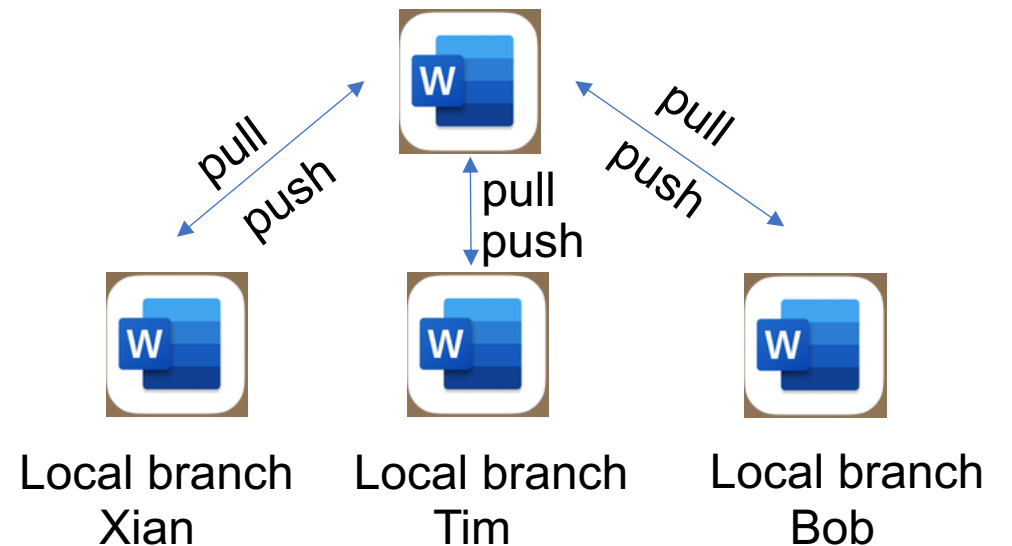
What is version control system and git?

VCS: **Tools** used to track changes to source code (or other collections of **files** and **folders**). (who wrote what at when + message...)

Ad-hoc (centralized) version control



Distributed version control



Git is the de facto standard for distributed version control.

Why we need git?

1. By yourself, small projects

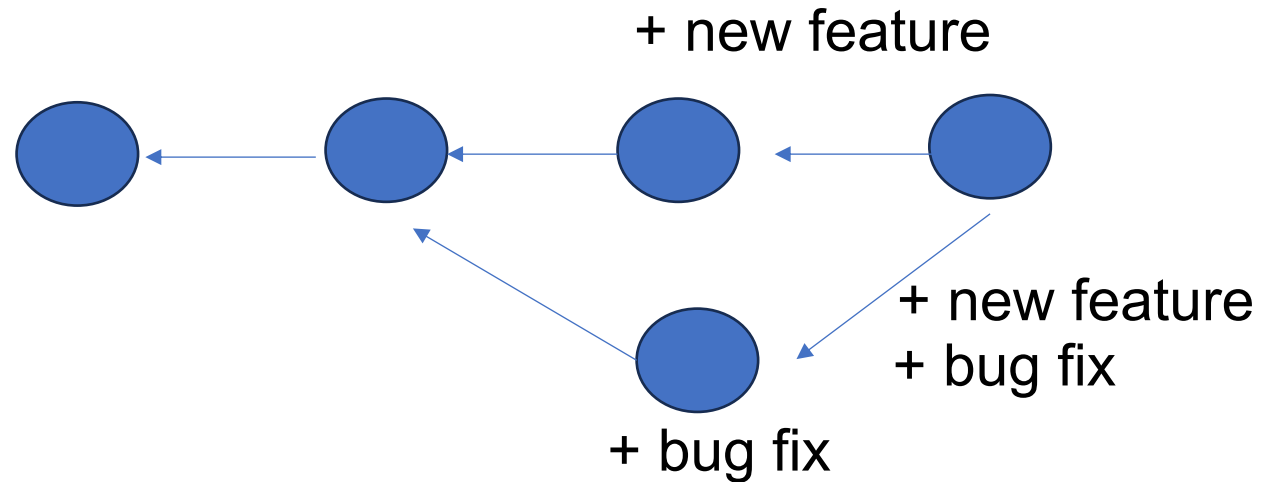
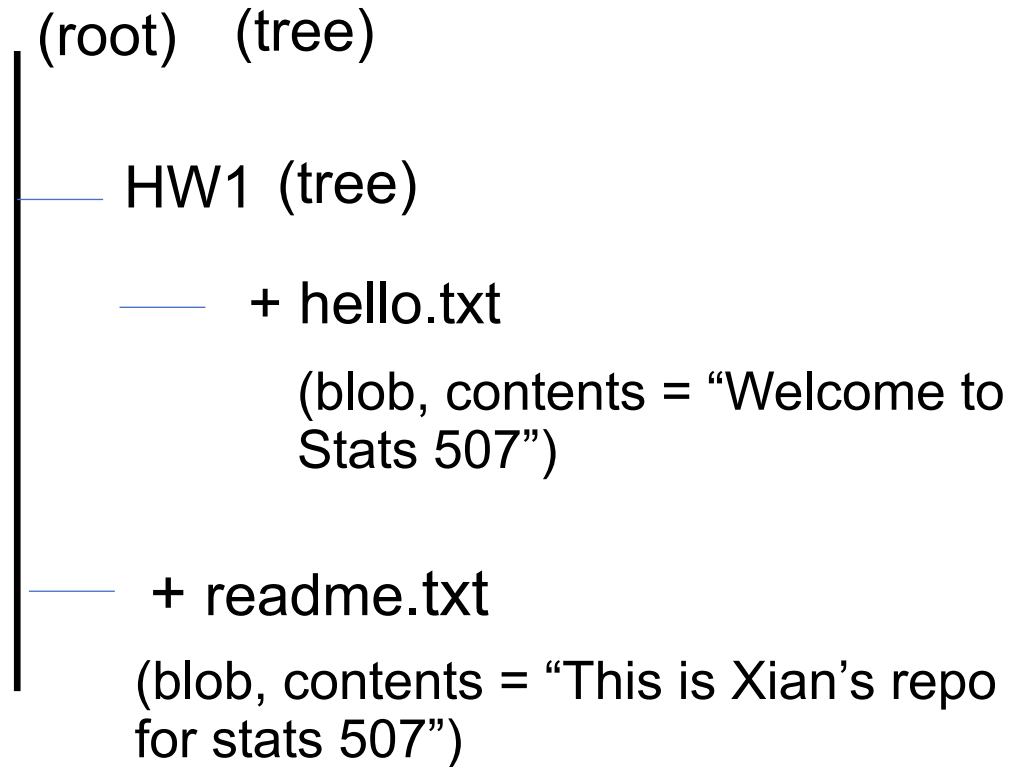
- Track history and keep a log of changes
- Parallel branches of development
- Build project portfolios (HWs and projects...)

2. Collaboration, larger projects

- Who wrote this module, when, why...
- Resolving conflicts
- Can easily roll back to previous versions (features were broken...)

How git works...

Git models the history of a collection of **files** and **folders** within some top-level directory as a series of **snapshots of history**.



Git commands

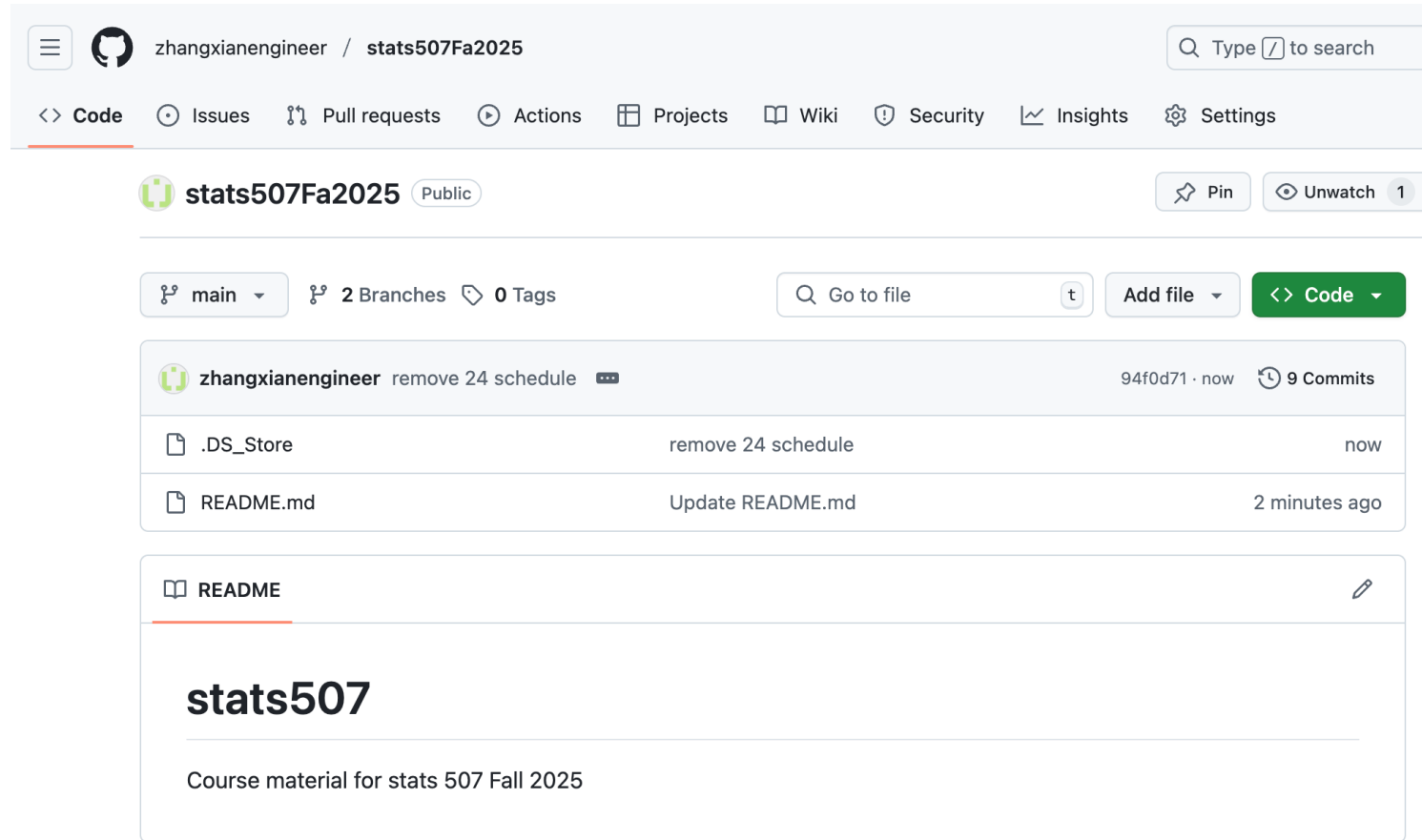
Basic Snapshotting

Command	Description
<code>git status</code>	Check status
<code>git add [file-name.txt]</code>	Add a file to the staging area
<code>git add -A</code>	Add all new and changed files to the staging area
<code>git commit -m "[commit message]"</code>	Commit changes
<code>git rm -r [file-name.txt]</code>	Remove a file (or folder)

More on: <https://github.com/joshnh/Git-Commands>

MIT missing semester version control lecture: <https://missing.csail.mit.edu/2020/version-control/>

Using github for STATS 507



The screenshot shows the GitHub interface for the repository 'stats507Fa2025' by user 'zhangxianengineer'. The repository is public and has 2 branches and 0 tags. The commit history shows two recent commits: 'remove 24 schedule' and 'Update README.md'. The README file is visible and contains the title 'stats507' and the description 'Course material for stats 507 Fall 2025'.

zhangxianengineer / stats507Fa2025

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

stats507Fa2025 Public Pin Unwatch 1

main 2 Branches 0 Tags Go to file Add file Code

zhangxianengineer remove 24 schedule 94f0d71 · now 9 Commits

File	Commit	Time
.DS_Store	remove 24 schedule	now
README.md	Update README.md	2 minutes ago

README

stats507

Course material for stats 507 Fall 2025

<https://github.com/zhangxianengineer/stats507Fa2025>

HW: sign up for [github](https://github.com) and create a stats 507 repo. You should **NOT** make your HWs public before the due date...

Week 1 practice problem

You will find them on Canvas in “Files/in-class practice”

First try it on your own, then we’ll discuss it.

Things to do:

Install [Anaconda](#) and [Jupyter](#)

Familiarize yourself with Jupyter:

<https://docs.jupyter.org/en/latest/start/index.html>

Sign up for [github](#) and create your own stats 507 repo.

Read ch1-6 in [Python 4 Everybody](#)

Other things

HW1 posted on canvas.

If you run into trouble, attend GSI office hours for help.

- You can also post to the canvas discussion board.
- If you're having trouble, at least one of your classmates is, too.
- You'll learn more by explaining things to teach other than by reading posts.